

Gauge-Invariant Scalarizability of Multi-Inverter Power Networks: Non-Abelian Cycle Dynamics and Connectivity–Stability Reversals

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Abstract—Scalar graph measures and decoupled impedance models are attractive for screening inverter-dominated power networks, but they discard matrix-valued feedback interactions created by phase-locked loops, voltage/current controls, grid-forming loops, and heterogeneous device dynamics. This paper develops a coordinate-invariant framework for deciding when such a network admits an exact scalar-channel representation, quantifying the error when it does not, and relating that model-validity question to connectivity–stability reversals. After port reduction, a frequency-dependent matrix return graph is normalized by a spanning-tree gauge and the non-tree transports become fundamental-cycle holonomies. We prove fixed-frequency scalarizability by simultaneous diagonalization and extend it to a broadband common-cycle criterion over a frequency band; noncommutativity and Jordan defectiveness are shown to be distinct exact obstructions. A physical energy metric then yields a gauge-invariant non-Abelian defect. More importantly, a spectral-gap theorem bounds the off-diagonal residual of every chord by its commutator with a reference holonomy. Across 2,000 structured 2×2 networks the bound has zero violations, a median slack of only 6.6%, and a 95th-percentile slack of 35.9%; the non-Abelian defect retains a strong association with scalarization error (Spearman $\rho = 0.8951$). A collective-cycle counterexample further shows that a global unit-return mode need not contain any individually resonant fundamental cycle, motivating gauge-invariant chord sensitivities rather than a single-loop margin. Finally, an AC-consistent IEEE 39-bus reduced SG/GFL/GFM study is used as a falsification-oriented engineering test. Under nominal PLL tuning, 180 candidate line additions produce no instability. In a deliberately low-damped PLL region, however, line 25–38 improves generalized-Fiedler connectivity while the full coupled mode crosses into instability and a decoupled model misses the event. The reversal occupies a finite line-strength/controller-gain region, disappears under well-damped GFL tuning and under the tested GFM controls, and is repaired by retuning one attributed PLL without removing the line. The resulting claim is deliberately narrow: non-Abelian cycle algebra is a certificate of scalar-model validity, not a stability metric; closed-loop stability must still be evaluated on the full matrix graph.

Index Terms—Inverter-dominated power systems, MIMO impedance, graph theory, holonomy, quiver representations, non-Abelian coupling, scalarization, generalized Fiedler vector, PLL, small-signal stability, Braess effect.

I. INTRODUCTION

MODERN power networks combine synchronous machines, grid-following (GFL) inverters, grid-forming (GFM) inverters, dynamic loads, and controls whose terminal dynamics are naturally multivariable. In this setting, a line is not merely a scalar weight connecting two otherwise identical oscillators. A perturbation can rotate, amplify, and mix voltage-angle and power channels as it traverses the network. Yet many planning and screening tools remain scalar: Laplacian connectivity, effective resistance, short-circuit-ratio families, or decoupled impedance approximations.

This mismatch matters for two reasons. First, the recently proposed generalized Fiedler vector (GFV) combines network connectivity with nodal inertia and provides a useful notion of *dynamic connectivity* [1]. Second, converter impedance studies have long shown that dropping coupling terms can lead to inaccurate eigenloci or even incorrect stability conclusions when mirror-frequency or dq coupling is material [6], [7]. These two facts create a practical gap: an upgrade can look favorable to a scalar connectivity metric while the closed-loop MIMO dynamics react in another direction.

Complex-frequency formulations provide an increasingly useful language for heterogeneous power-system dynamics [2], [3], [5]; however, complex frequency and GFV are not the contribution pursued here. Likewise, Braess-type stability degradation after adding a coupling is already known for oscillator networks and power grids [10]. The question addressed here is more specific:

When does a matrix-valued power-network model admit an exact scalar-channel representation, and what closed-loop information is missing when such a representation does not exist?

The answer developed in this paper uses cycle algebra. Each local interconnection is treated as a linear map between port spaces. Local state-coordinate changes act as gauge transformations. A spanning-tree gauge removes all tree transports, leaving the fundamental-cycle return maps—their *holonomies*—as the nontrivial global objects. This leads to four contributions.

C1. Exact scalarizability. A tree-gauge theorem characterizes when a connected matrix-valued network can be transformed into independent scalar channels: precisely when its fundamental-cycle holonomies are simultaneously diagonalizable.

- C2. Gauge-aware obstruction.** A positive-definite physical port metric yields a coordinate-invariant non-Abelian commutator defect. A 2,000-case structured Monte Carlo study shows that the defect tracks scalarization error, while commuting networks scalarize exactly to machine precision.
- C3. Collective feedback.** A counterexample shows that global unit-return resonance does not require any single fundamental cycle to be individually close to unit return. This motivates cycle-rooted/trace-based characteristic decompositions rather than a single “bad-loop” margin.
- C4. Connectivity–stability nonordering and mitigation.** An exact minimal MIMO counterexample and an AC-consistent IEEE 39-bus study show that improved scalar connectivity need not improve the full coupled spectrum. In the IEEE 39-bus stress case, a connectivity-improving line creates an instability missed by the decoupled model; targeted PLL retuning restores the margin without undoing the network upgrade.

The paper deliberately separates *scalarizability* from *stability*. Non-Abelianity is not proposed as a stability margin. Rather, it certifies when a scalar reduction cannot retain the complete matrix-valued feedback structure.

II. RELATED WORK AND POSITIONING

A. Scalar connectivity and complex frequency

The GFV recently proposed by Pinheiro and Dotta generalizes Fiedler connectivity by incorporating nodal inertia [1]. In parallel, local and system-wide complex-frequency formulations provide a device-agnostic description of voltage magnitude and angle dynamics [2], [3]. Büttner and Hellmann further introduced complex couplings as adaptive dynamical-network variables [5]. These developments provide useful baselines and future observables, but none addresses the exact reducibility of a heterogeneous matrix-valued interconnection into scalar channels.

B. Coupled versus decoupled converter models

Matrix impedance and generalized Nyquist approaches explicitly retain dq /sequence coupling. Rygg *et al.* showed that decoupled impedance models are exact only under specific coupling conditions and quantified eigenlocus error [6]. Xin *et al.* likewise showed that neglected coupling can produce incorrect conclusions and proposed a generalized-impedance treatment [7]. The present work is complementary: instead of assessing coupling at one converter, it asks whether the *entire network* admits one global basis in which every interconnection channel decouples.

C. Cycles, holonomy, and quiver determinants

Cycle-level topological constraints have recently appeared in synchronization through twisted Laplacians and $U(1)$ holonomy [9]. More generally, Kassel and Lévy derived determinant identities for linear quiver representations in terms of traces of cycle holonomies, including cycle-rooted spanning-forest interpretations for graph vector bundles [8]. We do not claim

these algebraic results. We use their perspective to motivate a power-system interpretation: matrix-valued feedback cycles are the coordinate-invariant structures left after trivializing a spanning tree.

D. Braess effects and stability nonordering

Adding a network coupling can either increase or decrease synchronous-state stability, including in power grids [10]. Our IEEE 39-bus experiment should therefore not be read as a claim that Braess behavior itself is new. The new target is the mismatch between scalar connectivity and a coupled converter mode, together with a scalarizability diagnosis and a controller-retuning remedy.

III. MATRIX-VALUED CLOSED-LOOP NETWORK

A. Port reduction and self-energy interpretation

Let z denote retained terminal variables and x_h internal machine, converter, filter, and controller states. Linearizing a DAE around an AC solution and eliminating x_h by a Schur complement yields a frequency-dependent port equation

$$F(s)z = 0, \quad (1)$$

where $F(s)$ contains the network and the dynamic self-energy of eliminated states. A convenient block splitting is

$$F(s) = D(s) - E(s), \quad (2)$$

with block-diagonal local dynamics $D(s) = \text{diag}(D_1, \dots, D_n)$ and off-diagonal interconnection blocks $E_{ij}(s)$. Whenever $D(s)$ is invertible,

$$F(s) = D(s)[I - T(s)], \quad T(s) = D^{-1}(s)E(s), \quad (3)$$

so

$$\det F(s) = \det D(s) \det [I - T(s)]. \quad (4)$$

A collective network mode therefore satisfies

$$1 \in \sigma[T(s)]. \quad (5)$$

The block $T_{ij}(s)$ maps the port space of node j to that of node i and embeds inertia, damping, controller memory, and network transfer. This is the point at which the earlier “feedback rheology” interpretation becomes useful: frequency-dependent local dynamics change the transport around network cycles rather than simply adding one scalar damping coefficient.

B. Gauge transformations

Let each port space be $V_i \simeq \mathbb{C}^d$ and let $x_i = Q_i \tilde{x}_i$ with $Q_i \in \text{GL}(d, \mathbb{C})$. Then

$$T_{ij} \mapsto \tilde{T}_{ij} = Q_i^{-1} T_{ij} Q_j. \quad (6)$$

The local realization can therefore change while the underlying physical interconnection does not. A useful network descriptor should either be invariant under (6) or carry the physical metric required to transform covariantly.

Definition 1 (Cycle holonomy). For an oriented closed walk $C = (i_0, i_1, \dots, i_r = i_0)$, define

$$H_C = T_{i_0 i_{r-1}} \cdots T_{i_2 i_1} T_{i_1 i_0}. \quad (7)$$

Under (6), $H_C \mapsto Q_{i_0}^{-1} H_C Q_{i_0}$. Hence its eigenvalues, trace, determinant, and characteristic polynomial are gauge-invariant.

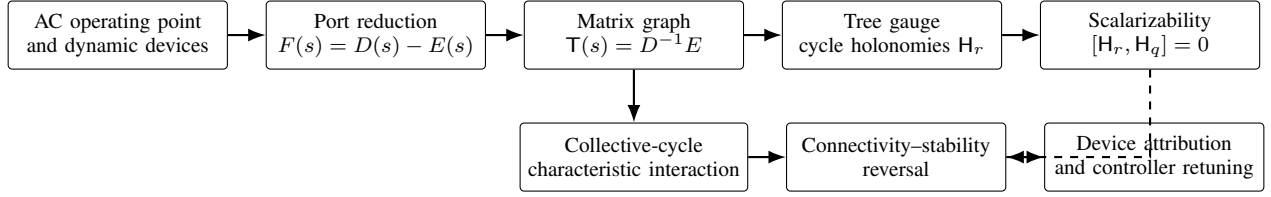


Fig. 1. Paper logic. The contribution is not a new scalar strength metric. The port-reduced MIMO network is treated as a matrix graph; cycle algebra decides exact scalarizability, while the full closed-loop model decides stability.

IV. EXACT SCALARIZABILITY

A. Single-cycle reduction and tree gauge

The simplest consistency check is a directed cycle with block transports $T_1, \dots, T_r$. Let T_C be the corresponding block cyclic return operator and $H_C = T_r \cdots T_1$.

Lemma 1 (Single-cycle determinant reduction). *For a pure directed cycle of equal block dimension,*

$$\det(I - T_C) = \det(I - H_C). \quad (8)$$

Proof. Successive block Gaussian elimination removes the tree-like chain blocks and leaves the Schur complement $I - T_r \cdots T_1$. Equivalently, the nonzero closed-walk contribution of the block permutation is the cycle product H_C . $\square$

A 3-node, 2×2 numerical check gives an absolute determinant mismatch of 3.5×10^{-16} . Under strongly non-unitary local coordinate changes, the spectrum, trace, and determinant of H_C remain invariant to machine precision. In contrast, the Euclidean quantity $\sigma_{\min}(I - H_C)$ changed from 0.9026 to 0.5132 under a pure coordinate transformation. This motivates the physical metric introduced below.

B. Tree gauge

Choose a spanning tree $\mathcal{T}_G$ and a root. Assume the tree transports are invertible. Recursively select the Q_i so that every tree transport becomes the identity. The remaining $\beta = m - n + 1$ chord matrices are exactly the fundamental-cycle holonomies in this tree trivialization.

Theorem 1 (Tree-gauge scalarizability). *For a connected matrix-valued network with equal port dimension d and invertible tree transports, the following statements are equivalent:*

- (i) *there exist local coordinate transformations that make every edge transport diagonal;*
- (ii) *in a tree gauge, all fundamental-cycle holonomies $H_1, \dots, H_\beta$ are simultaneously diagonalizable.*

If the H_r are individually diagonalizable over $\mathbb{C}$, (ii) is equivalent to pairwise commutation $[H_r, H_q] = 0$ for all r, q .

Proof. In a tree gauge all tree transports are I . Any further collection of local transformations that preserves all tree identities must satisfy $S_i = S_j$ on every tree edge; connectedness therefore implies a single similarity S at all vertices. The only nontrivial transports are the chord matrices H_r . Hence all edges can be diagonalized if and only if one S diagonalizes all H_r . The final statement is the standard simultaneous-diagonalization criterion for a commuting family of diagonalizable complex matrices [15]. $\square$

Remark 1. The theorem is a *structural* exactness result. It does not state that a commuting network is stable, only that it admits a representation as d independent scalar graph channels.

Because $T_{ij} = T_{ij}(s)$, pointwise scalarizability does not by itself guarantee one physically useful scalar decomposition over a frequency band. Let the spanning tree be fixed and, at each ω , perform the corresponding tree normalization. The remaining root-fiber holonomies are denoted $H_r(j\omega)$.

Definition 2 (Broadband common-cycle scalarizability). A matrix network is broadband common-cycle scalarizable on Ω if there exists one similarity S , independent of ω , such that $S^{-1}H_r(j\omega)S$ is diagonal for every fundamental cycle r and every $\omega \in \Omega$.

Theorem 2 (Broadband common-cycle criterion). *Assume that every member of the family $\{H_r(j\omega) : r = 1, \dots, \beta, \omega \in \Omega\}$ is diagonalizable over $\mathbb{C}$. Then broadband common-cycle scalarizability holds if and only if*

$$[H_r(j\omega), H_q(j\nu)] = 0 \quad \forall r, q, \omega, \nu \in \Omega. \quad (9)$$

Proof. The statement is simultaneous diagonalization of the entire frequency-indexed family. A simultaneously diagonal family commutes. Conversely, a commuting family of diagonalizable complex matrices is simultaneously diagonalizable [15]. $\square$

The diagonalizability hypothesis is essential. Pairwise commutation alone cannot detect a Jordan obstruction. For example,

$$J_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad J_2 = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \quad (10)$$

commute but neither is diagonalizable. We therefore distinguish the *non-Abelian obstruction* from the *Jordan obstruction*. One exact diagnostic is

$$\mathcal{J}(\omega) = \max_r \left[d - \sum_{\lambda \in \sigma(H_r)} \dim \ker(H_r - \lambda I) \right], \quad (11)$$

which vanishes exactly when every H_r is diagonalizable. Numerically, eigenvector conditioning should accompany $\mathcal{J}$ to detect near-defectiveness. Theorems 1–2 are structural statements, not stability conditions.

C. Gauge-invariant non-Abelian defect

A Euclidean Frobenius norm of $[H_r, H_q]$ is not invariant under non-unitary changes of local coordinates. Let $G \succ 0$ be

a physical quadratic port metric, transforming as $G \mapsto Q^* G Q$. Define

$$\|A\|_G^2 = \text{tr}(G^{-1} A^* G A) \quad (12)$$

and

$$\mathfrak{N}_{rq} = \frac{\|H_r H_q - H_q H_r\|_G}{\|H_r H_q\|_G + \|H_q H_r\|_G}. \quad (13)$$

Proposition 1 (Gauge invariance). *The norm (12) and defect (13) are invariant under $A \mapsto Q^{-1} A Q$, $G \mapsto Q^* G Q$.*

Proof. Substitution into (12), cyclicity of the trace, and $Q^{-1}(Q^*)^{-1} = (Q^* Q)^{-1}$ cancel all basis factors. Applying the same identity to the commutator and denominator terms yields (13) unchanged. $\square$

A strongly non-unitary numerical gauge transformation changed $\mathfrak{N}$ by 4.2×10^{-17} , whereas an unweighted Euclidean singular-value quantity changed materially. This correction is important: gauge invariance requires carrying the physical metric.

D. A quantitative spectral-gap certificate

Gauge-invariant commutators indicate whether exact scalarization is possible, but a screening tool also needs a quantitative statement. Let $G = W^* W \succ 0$ and whiten the physical metric by $\hat{H} = W H W^{-1}$. Choose a reference holonomy $\hat{H}_0$ with simple spectrum and write

$$S^{-1} \hat{H}_0 S = \Lambda = \text{diag}(\lambda_1, \dots, \lambda_d), \quad (14)$$

$$\delta_0 = \min_{i \neq j} |\lambda_i - \lambda_j| > 0. \quad (15)$$

For another holonomy $\hat{H}_q$, let $B_q = S^{-1} \hat{H}_q S$ and let $\text{off}(B_q)$ denote its off-diagonal part.

Theorem 3 (Commutator-to-scalarization bound). *Under the assumptions above,*

$$\|\text{off}(B_q)\|_F \leq \frac{\kappa_2(S)}{\delta_0} \|[\hat{H}_0, \hat{H}_q]\|_F. \quad (16)$$

Proof. In the reference eigenbasis, $[\Lambda, B_q]_{ij} = (\lambda_i - \lambda_j)(B_q)_{ij}$. Hence $\|\text{off}(B_q)\|_F \leq \delta_0^{-1} \|[\Lambda, B_q]\|_F$. Because $[\Lambda, B_q] = S^{-1}[\hat{H}_0, \hat{H}_q]S$, submultiplicativity gives (16). $\square$

The bound makes two failure mechanisms explicit: large cycle noncommutativity and a small reference spectral gap. The factor $\kappa_2(S)$ additionally penalizes a nearly defective reference basis. Thus the practical scalarization certificate combines commutator size, eigenvalue separation, and modal conditioning rather than treating $\mathfrak{N}$ alone as sufficient.

V. STRUCTURED MONTE CARLO SCALARIZATION TEST

A. Protocol

We generated 2,000 four-node networks with a directed spanning tree $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ and chords $2 \rightarrow 0$ and $3 \rightarrow 0$. Each 2×2 edge transport was a rotated anisotropic positive-definite matrix. After tree gauge and metric whitening, the first chord was used as the reference $\hat{H}_0$; the actual residual was the off-diagonal Frobenius norm of the second chord in the reference eigenbasis. The theorem bound in (16) was evaluated for every realization. In parallel, the full block network and the chord-diagonalized approximation were compared spectrally.

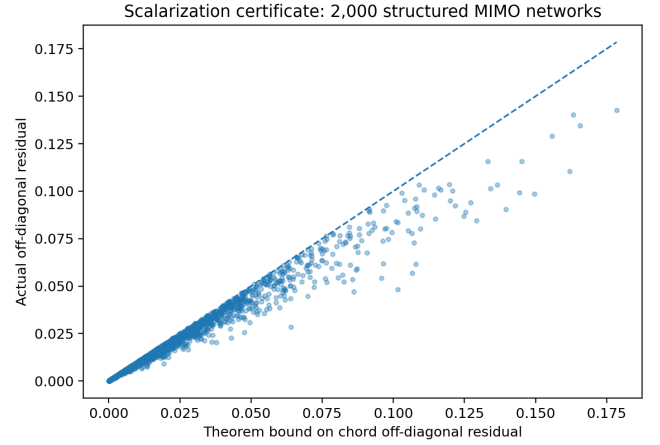


Fig. 2. Quantitative scalarization certificate for 2,000 structured MIMO networks. Every realization lies below the theorem line (dashed). The median bound/actual ratio is 1.066 and the 95th percentile is 1.359.

TABLE I
SCALARIZABILITY VALIDATION SUMMARY

Quantity	Result
Structured MIMO realizations	2000
Violations of (16)	0
Median bound/actual residual	1.0659
95th-pct. bound/actual residual	1.3589
Spearman $\rho_s(\mathfrak{N}, \epsilon_{sc})$	0.8951
Exactly commuting gauge tests	200
Max. commuting-case error	4.8×10^{-15}
Gauge-change error in $\mathfrak{N}$	4.2×10^{-17}

B. Results

There were no violations of (16) in 2,000 realizations. The median bound-to-actual ratio was 1.0659, and the 95th percentile was 1.3589. Thus, for this structured ensemble, the elementary spectral-gap certificate is substantially sharper than a subsequent Bauer–Fike bound on the full spectral shift, whose median slack exceeded $800\times$ and is therefore not used as the practical certificate.

The energy-metric non-Abelian defect remains informative about the final scalar-network error: Spearman $\rho_s = 0.8951$ and Pearson $r = 0.5306$. The upper-quartile median scalarization error is approximately 394 times the lower-quartile median. Finally, 200 exactly commuting cases under arbitrary local gauges remained exact to machine precision, with maximum scalarization error 4.8×10^{-15} .

The interpretation remains deliberately limited: the theorem and $\mathfrak{N}$ quantify *model reducibility*. They do not order closed-loop stability.

VI. COLLECTIVE-CYCLE FEEDBACK

A tempting cycle criterion would search for one fundamental cycle with $1 \in \sigma(H_C)$. The following counterexample shows why that is insufficient.

Proposition 2 (No single resonant cycle is necessary). *There exist matrix networks for which $1 \in \sigma(T)$ while $1 \notin \sigma(H_r)$ for every fundamental-cycle holonomy.*

Constructive numerical proof. In a four-port, two-chord tree gauge, the global return operator satisfies $1 \in \sigma(T)$ to machine precision. However,

$$\sigma(H_1) = \{0.51884, 0.24877\}, \quad (17)$$

$$\sigma(H_2) = \{0.50223, 0.18559\}, \quad (18)$$

so the nearest individual cycle is more than 0.48 from unit return. Nevertheless, sensitivities of the global unit eigenvalue to the two chord gains are 0.14528 and 0.14104, respectively. Thus both cycles participate in the global condition although neither is individually resonant. $\square$

This motivates a *collective-cycle* characteristic description. Recent trace identities for quiver representations express suitable block determinants through traces of products along cycles, and twisted-Laplacian determinants through cycle-rooted spanning forests [8]. We use this machinery as mathematical infrastructure rather than claim it as new.

A directly useful power-system quantity follows from eigenvalue perturbation of the return graph. Suppose $T(s_k)x_k = x_k$, $y_k^* T(s_k) = y_k^*$ and $y_k^* x_k = 1$. Let a tree-gauged chord block H_a be scaled by a scalar γ_a .

Proposition 3 (Gauge-invariant chord sensitivity). *For a simple unit-return eigenvalue,*

$$\frac{d\lambda_T}{d\gamma_a} = y_{k,i}^* H_a x_{k,j}, \quad (19)$$

and the corresponding closed-loop mode sensitivity is

$$\frac{ds_k}{d\gamma_a} = -\frac{y_{k,i}^* H_a x_{k,j}}{y_k^* T_s(s_k) x_k}. \quad (20)$$

Both expressions are invariant under local gauge changes when the left/right vectors transform contragrediently.

Proof. Differentiate $y_k^*(T - \lambda_T I)x_k = 0$ with respect to γ_a to obtain (19). At a closed-loop mode $\lambda_T(s_k) = 1$, implicit differentiation of $\lambda_T(s_k, \gamma_a) = 1$ yields (20). $\square$

For a complete characteristic attribution, Kassel–Lévy-type trace/CRSF identities motivate

$$P(s) = \det F(s) = \sum_{\mathcal{F}} \Psi_{\mathcal{F}}(s), \quad (21)$$

but Proposition 3 already provides a scalable chord-level diagnostic without enumerating all cycle-rooted spanning forests.

VII. CONNECTIVITY DOES NOT ORDER MIMO STABILITY

A. Exact minimal counterexample

Consider

$$\dot{x} = (I \otimes A - L \otimes C)x \quad (22)$$

for two nodes joined by an edge of weight w , so $\mu_2(L) = 2w$. Let

$$A = \begin{bmatrix} 0.719614 & -1.819392 \\ 0.794706 & -1.686641 \end{bmatrix}, \quad (23)$$

$$C = \begin{bmatrix} 0.283668 & -0.501948 \\ 0.467149 & 0.202568 \end{bmatrix}.$$

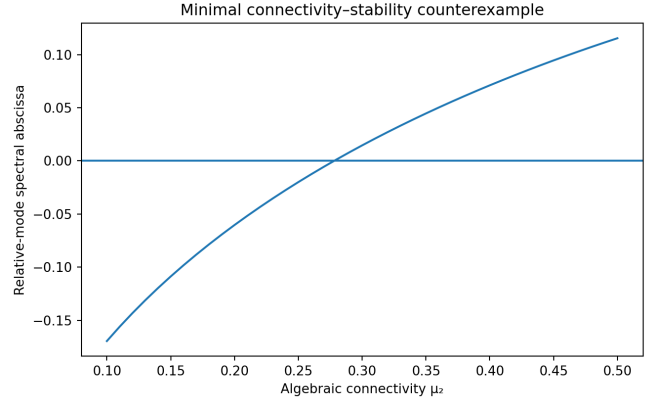


Fig. 3. Exact minimal counterexample. Increasing algebraic connectivity crosses the closed-loop stability boundary at $\mu_c = 0.2780313$. The purpose is to disprove a universal connectivity ordering, not to claim a new diffusion-driven instability mechanism.

Both A and $-C$ are Hurwitz. The relative mode is $A - \mu C$ and

$$\det(A - \mu C) = 0.2919466\mu^2 - 0.9161545\mu + 0.2321517. \quad (24)$$

The two positive roots are

$$\mu_c = 0.2780313, \quad \mu_2^{(+)} = 2.8600578, \quad (25)$$

or, in edge weight,

$$w_c = 0.1390156, \quad w_2 = 1.4300289. \quad (26)$$

The trace remains negative, while the determinant is negative between the two roots. Hence one relative-mode eigenvalue is positive for $w_c < w < w_2$. The first crossing at w_c is the connectivity-increase destabilization used in the argument.

Theorem 4 (Connectivity–stability nonordering). *There is no universal ordering in which increasing scalar algebraic connectivity μ_2 must improve the spectral abscissa of a matrix-coupled network. In particular, the system (22)–(23) is stable for $0 < w < w_c$ and unstable on the nonempty interval $w_c < w < w_2$, with $w_2 = 1.4300289$, although $\mu_2 = 2w$ increases monotonically throughout the interval.*

Related nonmonotone phenomena are known from master-stability and diffusion-driven systems [11]; the contribution here is to use the counterexample as the minimal analytical support for the converter-network screening problem studied next.

VIII. AC-CONSISTENT IEEE 39-BUS VALIDATION

A. Electrical benchmark and device placement

The solved MATPOWER New England 39-bus case is used as the base AC operating point [13]. For every candidate new line, the nonlinear AC power flow is re-solved, the full P, Q injection Jacobian with respect to (θ, V) is recomputed, and the 29 load buses are eliminated by a Schur complement. The remaining ten ports correspond to the generator buses.

Four converter placements follow the commonly used low-inertia Config.-II replacement pattern, corresponding here to

TABLE II
IEEE 39-BUS DYNAMIC SCREENING CASES

Case	PLL K_p	PLL K_i
Nominal	25	400
Low-damped stress	1	400
Mitigated bus 30	3 (bus 30), 1 otherwise	400

buses 30, 34, 37, and 38. The public EPFL benchmark provides full-replica configurations in which four synchronous machines are replaced by type-III wind plants [14]. Our present device equations are reduced surrogates rather than the full EPFL Simulink models; this distinction is maintained throughout the paper.

B. Reduced dynamic models

The synchronous generators use a classical source-behind-reactance representation with swing dynamics. Each GFL device uses a PI PLL and first-order active/reactive current-command surrogates while retaining first-order P/Q cross-coupling caused by PLL angle error. A filtered $P-f/Q-V$ GFM helper model is included in the reproducibility code, although the strongest reversal below uses GFL units. The resulting reduced model is physically structured but should not be confused with a vendor-grade EMT realization. The retained core dynamics are, schematically,

$$\dot{\delta}_g = \omega_b \Delta\omega_g, \quad (27)$$

$$M_g \dot{\Delta\omega}_g = -D_g \Delta\omega_g - \Delta P_{e,g}, \quad (28)$$

for each SG, and

$$\dot{\hat{\theta}}_i = -K_{p,i} V_i (\hat{\theta}_i - \theta_i) + K_{i,i} \xi_i, \quad (29)$$

$$\dot{\xi}_i = -V_i (\hat{\theta}_i - \theta_i), \quad (30)$$

for each GFL PLL, augmented by first-order P/Q command states. Linearizing the terminal rotation gives cross terms proportional to the operating-point P_i and Q_i ; these are retained in the full model and removed in the deliberately decoupled baseline. A GFM helper uses filtered $P-f$ and $Q-V$ droop states.

For graph baselines, a synchronizing Laplacian

$$K = \sum_{(i,j)} V_i V_j B_{ij} \cos(\delta_i - \delta_j) b_{ij} b_{ij}^T \quad (31)$$

is Kron-reduced to the ten retained ports. Standard Fiedler connectivity is the second eigenvalue of K_r . GFV connectivity is the second generalized eigenvalue

$$K_r \phi = \mu M \phi, \quad (32)$$

with synchronous-machine inertia and small converter inertia surrogates.

C. Nominal-tuning falsification gate

With $K_p = 25$, $K_i = 400$, the baseline critical mode is

$$-0.15503 + j7.07240 \text{ s}^{-1}. \quad (33)$$

Among 180 candidate line additions at graph distance two or three, 173 increase GFV and 73 both increase GFV and move the critical real part to the right. None becomes unstable. This is an important negative result: the paper does not claim that ordinary line reinforcement generically destabilizes a well-tuned converter network.

D. Low-damped PLL stress gate

We next set $K_p = 1$, $K_i = 400$ on the four GFL units. In the local normalized PLL surrogate this corresponds to approximately $\omega_n = 20.5 \text{ rad/s}$ (3.26 Hz) and $\zeta \approx 0.026$. The baseline is still stable:

$$s_0 = -0.01507 + j17.22591 \text{ s}^{-1}. \quad (34)$$

Of the same 180 candidate lines, four both increase GFV and produce an unstable full coupled mode. Table III lists them.

IX. CASE STUDY: LINE 25–38

A. Connectivity improves while the coupled mode destabilizes

For the full line addition,

$$\mu_2^{\text{GFV}} : 0.2349169 \rightarrow 0.2352597 \quad (+0.1459\%), \quad (35)$$

$$\lambda_2(K_r) : 12.08886 \rightarrow 12.12489 \quad (+0.2980\%). \quad (36)$$

At the same time, the full coupled critical mode moves to

$$s_{\text{crit}} = +0.25019 + j19.77892 \text{ s}^{-1}. \quad (37)$$

The stability crossing occurs already at $\alpha_c = 0.21298$ of the candidate line admittance; at the crossing, GFV remains 0.1361% above its pre-upgrade value and standard Fiedler 0.3356% above its pre-upgrade value.

B. Decoupled model misses the event

A decoupled approximation removes the $P-V$, $Q-\theta$, and PLL-induced P/Q rotation-coupling terms. It predicts

$$-0.15627 + j7.30340 \quad \text{before the line}, \quad (38)$$

$$-0.15625 + j7.30515 \quad \text{after the line}. \quad (39)$$

Thus the scalarized/decoupled representation indicates essentially no adverse change while the full coupled model moves from stable to unstable. This is the main engineering use case for the scalarizability viewpoint: the error is not that a scalar graph metric is numerically poor, but that a scalar representation has omitted a coupled feedback channel.

C. Cycle-algebra diagnosis of the IEEE 39-bus mode

The converter-port model was next evaluated directly as the return graph $T(s) = D^{-1}(s)E(s)$. A maximum-weight generator-port spanning tree was fixed and all remaining directed port couplings were treated as chord closures in the tree gauge. A well-conditioned reference holonomy was used to construct the best compatible cycle-scalarized basis in the PLL band.

At the exact full-model modes, the cycle off-diagonal residual changes from 0.0673 before the upgrade to 0.1733 at the line-strength stability crossing and 0.1560 after the full line.

IEEE 39-bus study topology and converter placement

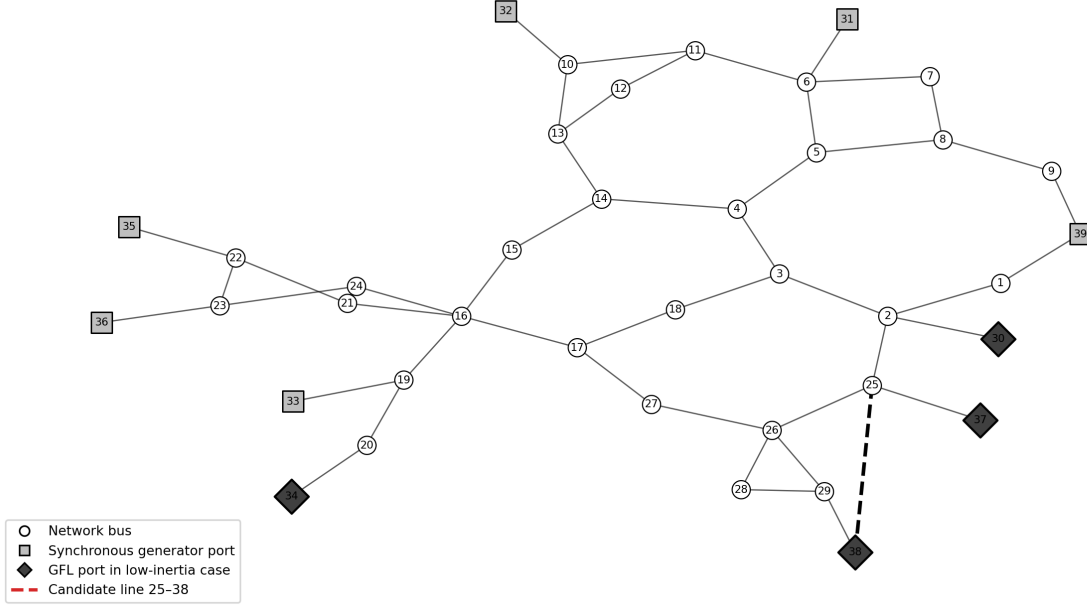


Fig. 4. IEEE 39-bus study topology. Generator buses are retained as dynamic ports after load-bus Schur reduction. The low-inertia GFL placements are buses 30, 34, 37, and 38; line 25–38 is the principal candidate intervention. The full benchmark topology and operating point are taken from MATPOWER, while converter dynamics are reduced surrogates.

TABLE III
CONNECTIVITY-IMPROVING LINE ADDITIONS THAT DESTABILIZE THE LOW-DAMPED PLL STRESS CASE

Line	Graph distance	Δ Fiedler (%)	Δ GFV (%)	New real part (s^{-1})	Imag. part (s^{-1})	Interpretation
25–38	3	0.298	0.146	0.2502	19.7789	strongest reversal
17–29	3	4.419	0.740	0.0963	19.8752	unstable
2–29	3	0.381	0.952	0.0252	19.6046	unstable
4–30	3	2.013	1.020	0.00057	19.7921	near boundary

More importantly, the full return graph satisfies the unit-return condition to 5.4×10^{-15} at the crossing, while the cycle-scalarized model remains 0.296 away from unit return. Thus the scalarization failure is not only visible as a generic commutator: it prevents the reduced cycle channels from reproducing the actual closed-loop mode.

The gain-weighted aggregate non-Abelian defect over all chord pairs rises from 0.0452 before the line to 0.0567 at the crossing and 0.0785 after the full line. It becomes 0.0823 after stabilizing retuning; therefore it is *not* a stability margin. Its role is to certify that the coupled cycle algebra cannot be faithfully replaced by independent scalar loops.

Proposition 3 localizes that collective interaction. At the post-upgrade unstable mode the largest gauge-invariant chord closure contributions are shown in Table IV. Several converter-related closures contribute simultaneously rather than one isolated “bad cycle.”

Finally, the broadband criterion is not satisfied in the PLL band. For example, the same effective cycle closure associated with the 34–33 port path evaluated at 15 and 25 rad/s has a nonzero cross-frequency commutator; a normalized energy-metric witness is approximately 0.71. Hence there is no single

TABLE IV
LARGEST POST-UPGRADE EFFECTIVE CHORD CONTRIBUTIONS

Source	Target	$ y_t^* H_a x_j $
38	37	0.0822
30	38	0.0731
38	30	0.0727
37	30	0.0501
34	33	0.0434
39	30	0.0358
38	39	0.0273
39	38	0.0260

common cycle basis that diagonalizes the relevant dynamic transports across that band.

X. MODE ATTRIBUTION AND CONTROLLER RETUNING

Biorthogonal participation of the unstable mode is concentrated in the four GFL units, as shown in Fig. 7a. The largest magnitudes are 0.573 at bus 30, 0.380 at bus 37, 0.236 at bus 38, and 0.084 at bus 34. The endpoint bus 38 participates, but the dominant contributor is remote bus 30. This supports the collective-feedback interpretation.

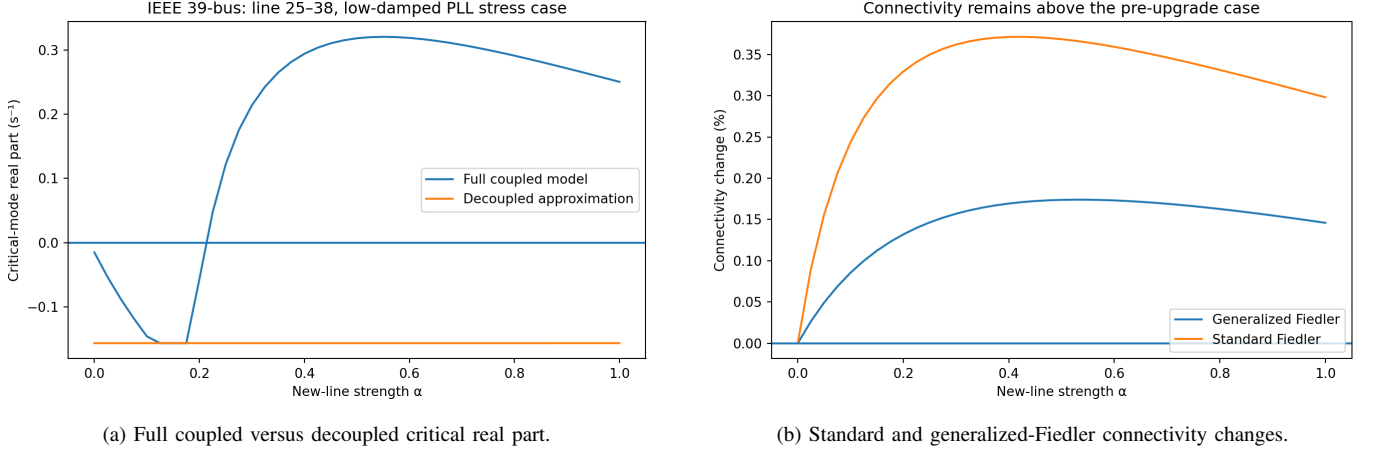


Fig. 5. IEEE 39-bus line 25–38 stress-case sweep. Connectivity remains above the pre-upgrade case while the full coupled model crosses into instability. The decoupled approximation does not reveal the dangerous mode.

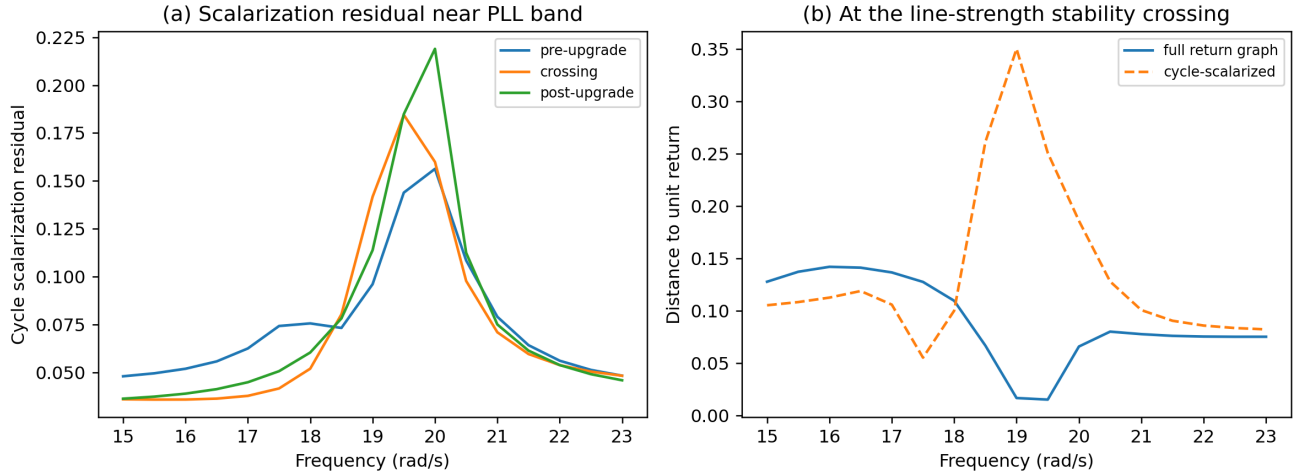


Fig. 6. Cycle-algebra gate for the IEEE 39-bus line 25–38 case. (a) The compatible cycle-scalarization residual grows in the PLL band as the line approaches the stability crossing. (b) At the crossing the full return graph approaches unit return, whereas the cycle-scalarized representation remains far away.

Retuning only the proportional PLL gain at bus 30 gives a zero crossing at

$$K_{p,30}^* = 2.10837. \quad (40)$$

With $K_{p,30} = 3$, the new-line case has

$$s_{crit} = -0.10064 + j19.62628 \text{ s}^{-1}, \quad (41)$$

while the improved graph connectivity is unchanged. The practical workflow is therefore

$$\begin{aligned} \text{upgrade} &\rightarrow \text{coupled-mode diagnosis} \\ &\rightarrow \text{device attribution} \rightarrow \text{retuning}. \end{aligned} \quad (42)$$

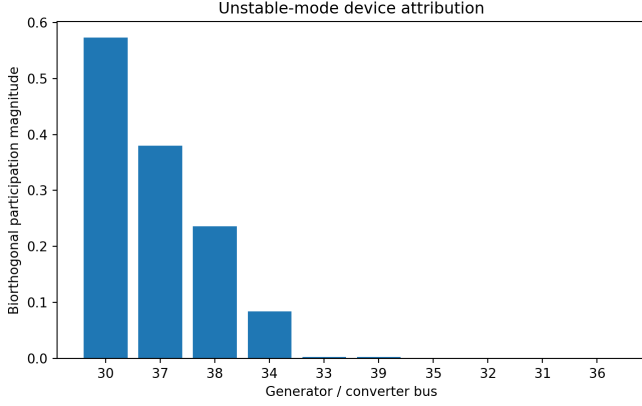
A. The reversal is a finite, controller-specific region

The stress result is not an isolated point in parameter space. Figure 8 sweeps the line-strength factor $\alpha \in [0, 1]$ and a common GFL PLL proportional gain K_p for $K_i = 400$. The zero contour encloses a finite unstable region. The pre-line stability boundary is $K_p \approx 0.959$, whereas with the complete

25–38 line it moves to $K_p \approx 1.513$. Thus $K_p = 1$ is stable before the line and unstable after a sufficiently strong line; even $K_p = 1.5$ is baseline-stable but gives a small positive post-line real part ($+0.00629 \text{ s}^{-1}$). For $K_p \geq 2$ the tested GFL model remains stable.

The boundary persists when K_i is changed. At $\alpha = 1$, the minimum stable common K_p is approximately 1.253, 1.513, and 1.741 for $K_i = 300, 400$, and 500, respectively. This establishes a nonzero parameter region rather than a single hand-picked tuning.

A negative control is equally important. Replacing the four converter ports by the tested GFM P - f/Q - V surrogate gives stable pre/post-line modes for droop values $R_p = 0.02, 0.04$, and 0.08, despite the same connectivity increase. Similarly, well-damped GFL tuning ($K_p = 25$) remains stable. The reversal is therefore *controller-specific*; it is not a topological claim that adding line 25–38 is inherently destabilizing.



(a) Biorthogonal device participation of the unstable mode.

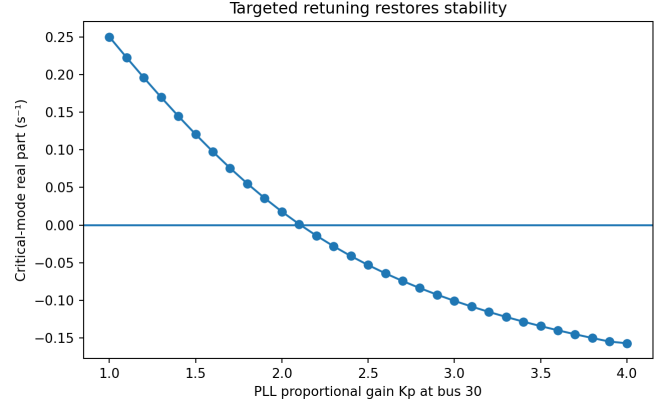
(b) Retuning only $K_{p,30}$ restores the stability margin.

Fig. 7. Diagnosis and mitigation. The line upgrade is retained; only the dominant converter controller is retuned.

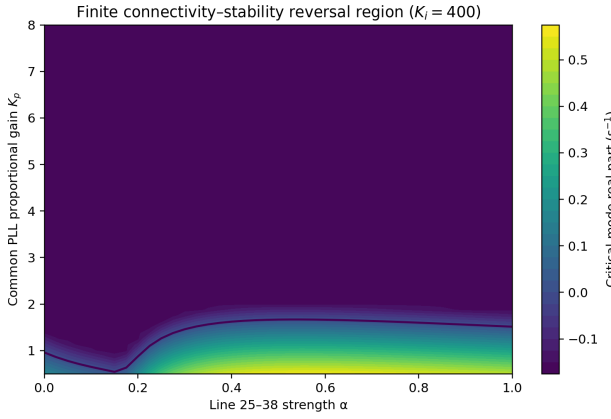
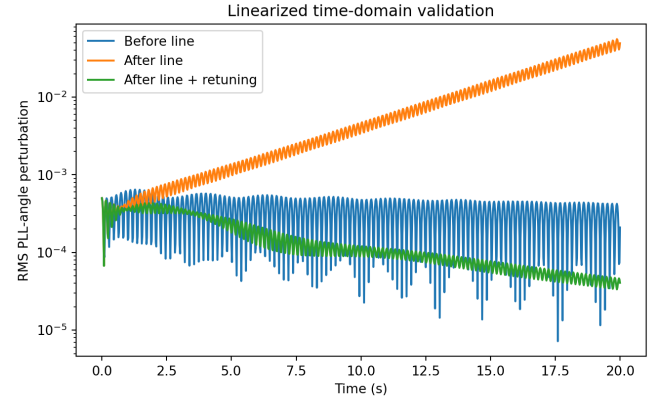
Fig. 8. Finite connectivity-stability reversal region for $K_i = 400$. The contour is the full coupled stability boundary. Connectivity remains above the pre-upgrade case as α grows; the instability is controlled by the closed-loop device/network interaction.

Fig. 9. Linearized time-domain validation of the line 25–38 stress case. The new-line response grows without retuning and decays after targeted PLL retuning.

B. Time-domain consistency

A small localized PLL-angle perturbation at bus 38 was propagated through the linearized system. Over 20 s, the unretuned new-line RMS PLL-angle perturbation grows from approximately 5×10^{-4} to 4.93×10^{-2} , whereas the retuned case decays to approximately 4.1×10^{-5} .

XI. INTERPRETATION: WHAT IS NEW AND WHAT IS NOT

A. What the results establish

The experiments and proofs support four statements.

- (1) Exact pointwise scalarization is controlled by simultaneous diagonalizability of fundamental-cycle holonomies, while a common scalar basis over a band requires simultaneous diagonalization of the full frequency-indexed family. Noncommutativity and Jordan defectiveness are distinct obstructions.
- (2) A physically metric-compatible commutator is coordinate invariant, and Theorem 3 turns the commutator into a quantitative off-diagonal error certificate with a spectral-gap/conditioning penalty.

- (3) Global resonance can be collective across cycles. Proposition 3 provides a gauge-invariant chord sensitivity, and the IEEE 39-bus case exhibits several simultaneous converter-related cycle closures rather than one isolated resonant loop.
- (4) Scalar connectivity and the coupled closed-loop spectrum are not ordered in general. The IEEE reversal occupies a finite controller/line-strength region, is absent under nominal GFL tuning and the tested GFM controls, and can be repaired by targeted retuning without undoing the connectivity upgrade.

B. What the results do not establish

The non-Abelian defect $\mathfrak{N}$ is not claimed to be a stability margin. High $\mathfrak{N}$ means that a scalar representation is structurally questionable, not that the system is necessarily unstable. The nominal IEEE 39-bus scan found no unstable line addition. The reversal appears in a deliberately low-damped PLL stress case, which is exactly where coupled modeling should matter most.

The AC network and operating-point re-solution are benchmark-consistent, but the GFL/GFM dynamic blocks are reduced surrogates. Therefore the IEEE 39-bus result is a constructive AC-consistent proof-of-concept, not yet a full-fidelity industry validation.

C. Relation to complex frequency

Sofos and Milano showed that complex frequency can coincide with system eigenvalues for diagonalizable LTI systems under the appropriate transformed states, while nonlinear trajectories generally break this equivalence [3]. This gives a natural future observation layer: the coupled mode predicted by the matrix graph may be tracked through complex-frequency patterns measured at converter terminals. That measurement connection is not required for the theorems above and is intentionally left outside the contribution claim.

XII. ENGINEERING WORKFLOW

The proposed use of the framework is not to replace full MIMO stability analysis. It is to decide when a reduced scalar analysis is defensible and, when it is not, to preserve the smallest coupled model required for diagnosis.

- Step 1.** Solve the AC operating point and obtain a local port model $F(s)$, either analytically, by frequency response, or by DAE elimination.
- Step 2.** Construct the matrix return graph $T(s)$ and a fundamental-cycle basis.
- Step 3.** Carry a physical port metric, whiten the cycle matrices, and evaluate commutators, Jordan/eigenvector conditioning, and the bound (16) over the frequency band of interest.
- Step 4.** If the holonomy family is diagonalizable and commuting at the required frequency resolution, use the exact or bounded near-scalar channels. If a common broadband basis is required, also test the cross-frequency condition (9); otherwise retain the coupled model.
- Step 5.** Compare graph planning metrics such as GFV with the full coupled spectral margin; do not infer one from the other.
- Step 6.** If an intervention creates a dangerous coupled mode, use left/right participation and, ultimately, cycle-rooted attribution to identify the controller-network structures that move it.
- Step 7.** Retune the smallest responsible controller set and verify the full coupled and time-domain response.

XIII. LIMITATIONS AND FINAL VALIDATION GATE

The remaining external validation gate is one full-order replication. The public EPFL IEEE 39-bus benchmark explicitly provides full-replica Config. I, low-inertia Config. II (four synchronous machines replaced by type-III wind plants), a BESS/VSC variant, and very-low-inertia Config. III [14]. Those models require the MATLAB/Simulink/ARTEMIS environment and cannot be executed in the present reproducibility runtime; alternatively, an independently implemented ANDES/EMT model can be used. The required gate is narrow:

- (i) reproduce a connectivity-improving network intervention in a full DAE/EMT model;
- (ii) compare full MIMO and decoupled analysis;
- (iii) confirm the critical mode in time domain;
- (iv) retune the highest-participation controller and recover the margin without undoing the network upgrade.

A successful full-order reversal would elevate the current proof-of-concept into a strong validation of the proposed screening logic. A failure to reproduce it would still leave the scalarizability theorem and nonordering counterexample intact, while narrowing the engineering claim.

XIV. CONCLUSION

This paper separates model reducibility from power-system stability. After port reduction, a multi-inverter network becomes a frequency-dependent matrix graph whose fundamental-cycle holonomies contain the global coordinate-invariant obstruction to scalar channels. Exact pointwise and broadband scalarizability reduce to simultaneous diagonalization, with noncommutativity and Jordan defectiveness as separate obstructions. A spectral-gap theorem converts the cycle commutator into a quantitative bound on the chord off-diagonal residual; in 2,000 structured networks the bound has zero violations and modest slack. Global unit-return resonance is nevertheless collective: no single fundamental cycle need be resonant, and gauge-invariant chord sensitivities expose how multiple closures contribute to a mode.

The IEEE 39-bus reduced study then supplies a deliberately falsification-oriented engineering example. Connectivity improves under line 25–38, yet a low-damped GFL region contains a finite parameter set in which the full coupled mode crosses into instability while a decoupled model misses the event. The new cycle diagnostics show that the same mode cannot be reconstructed by independent scalar cycle channels, and chord sensitivity exposes a distributed converter-related feedback structure. Well-damped GFL tuning and the tested GFM controls remain stable, while targeted PLL retuning repairs the stress case without removing the line. The conclusion is therefore narrower than a new strength metric: *non-Abelian cycle algebra certifies when scalar analysis is structurally inadequate; stability remains a property of the full closed-loop matrix graph.*

APPENDIX A

BROADBAND AND JORDAN DETAILS

Theorem 2 is a statement about one common similarity on the root fiber after frequency-dependent tree normalization. It should not be confused with a frequency-independent local state transformation in the original node coordinates. The distinction matters because the tree transport itself can be dispersive. In computations, a sampled broadband obstruction can be formed from cross-frequency commutators of the whitened holonomy family. A zero commutator family must still be checked for diagonalizability; the exact deficiency (11) and eigenvector conditioning provide complementary tests.

APPENDIX B PROOF DETAILS FOR THEOREM 3

Let $B_q = S^{-1}\hat{H}_q S$. Since Λ is diagonal,

$$\|[\Lambda, B_q]\|_F^2 = \sum_{i \neq j} |\lambda_i - \lambda_j|^2 |(B_q)_{ij}|^2 \geq \delta_0^2 \|\text{off}(B_q)\|_F^2. \quad (43)$$

Moreover,

$$[\Lambda, B_q] = S^{-1}[\hat{H}_0, \hat{H}_q]S, \quad (44)$$

so $\|[\Lambda, B_q]\|_F \leq \|S^{-1}\|_2 \|S\|_2 \|\hat{H}_0, \hat{H}_q\|_F$. Combining the inequalities proves (16). The whitening construction is gauge compatible: if $G = W^*W$ and $H' = Q^{-1}HQ$, $G' = Q^*GQ$, choosing $W' = WQ$ yields $W'H'(W')^{-1} = WHW^{-1}$.

APPENDIX C ADDITIONAL PROOF DETAILS FOR THEOREM 1

Let the spanning tree be rooted at r . Set $Q_r = I$. For every tree edge $i \rightarrow j$, define $Q_j = T_{ji}Q_i$; if the stored orientation is opposite to the tree traversal, use the inverse transport. Then

$$Q_j^{-1}T_{ji}Q_i = I. \quad (45)$$

Every chord $a : u \rightarrow v$ becomes $Q_v^{-1}T_a Q_u$, which is the holonomy obtained by traversing the tree from r to u , then the chord, then the inverse tree path from v to r . If a second gauge S_i leaves every tree edge equal to I , then $S_j^{-1}S_i = I$ on every tree edge; connectedness implies $S_i = S$ for all i . Consequently all chords can be made diagonal exactly when one common S diagonalizes them. This proves the equivalence in Theorem 1.

APPENDIX D COUNTEREXAMPLE STABILITY THRESHOLD

For (23), the relative-mode trace is

$$\text{tr}(A - \mu C) = \text{tr}(A) - \mu \text{tr}(C), \quad (46)$$

which remains negative at the first positive determinant root. Hence the two-dimensional Routh–Hurwitz conditions reduce the first loss of stability to $\det(A - \mu C) = 0$, yielding (26). This makes the counterexample exact rather than simulation-dependent.

APPENDIX E COLLECTIVE-CYCLE COUNTEREXAMPLE DATA

For Proposition 2, the two chord holonomies in tree gauge are

$$\begin{aligned} H_1 &= \begin{bmatrix} 0.455247 & 0.113039 \\ 0.116163 & 0.312369 \end{bmatrix}, \\ H_2 &= \begin{bmatrix} 0.498997 & 0.057555 \\ 0.017625 & 0.188825 \end{bmatrix}. \end{aligned} \quad (47)$$

Their energy-metric commutator defect for $G = I$ is 0.08588. The full 8×8 return operator is scaled so that its dominant eigenvalue is exactly one; the complete matrix is included in the reproducibility package. The important point is structural: the global unit-return condition emerges without either H_1 or H_2 approaching one individually.

APPENDIX F REDUCED DEVICE-MODEL SCOPE

The IEEE 39-bus surrogate intentionally retains only the dynamic mechanisms required to test the paper hypothesis. SG inertia/damping is represented explicitly; GFL devices retain PLL states and the operating-point rotation that mixes active/reactive injection channels; line additions trigger a new nonlinear AC solution before each linearization. Current limits, dc-link energy, detailed electromagnetic filters, saturation, protection logic, and vendor-specific inner loops are omitted. These omissions define the final full-order validation gate rather than being hidden assumptions.

APPENDIX G REPRODUCIBILITY SUMMARY

The numerical package accompanying this draft contains: (i) the 2,000-case scalarizability Monte Carlo script and CSV; (ii) the analytical counterexample script and sweep; (iii) the MATPOWER case39 data; (iv) the AC power-flow, Schur-reduction, SG/GFL/GFM surrogate, line-screening, participation, retuning, and time-response scripts; and (v) all figure data. The random seed for the scalarizability campaign is 20260823.

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